Geoheritage (2025) 17:34
https://doi.org/10.1007/s12371-025-01076-7

ORIGINAL ARTICLE

# Enhancing Geotourism in Southeastern Morocco through Machine Learning-Based Geomorphosite Identification

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Received: 23 March 2024 / Accepted: 28 January 2025 / Published online: 14 February 2025


**Abstract**

Mapping optimal areas for finding geomorphosites in large areas is a complex task influenced by various factors. To address this challenge, the present study assesses the effectiveness of three widely used machine learning classifiers in identifying and mapping potential areas for geomorphosite, a crucial factor for attracting and enhancing geotourism in Ziz, southeast Morocco. Initially, a comprehensive inventory of 120 geomorphosites was conducted in the study area. At each site, precise measurements of three topographical parameters were taken. Following this, three machine learning algorithms, namely Random Forest, Multi-Layer Perceptron, and M5 Prime, were utilized to create predictive models. Regarding the model performance, the Multi-Layer Perceptron model achieved the highest performance with an area under the curve of 0.91, followed by the M5 Prime model with 0.77. These models identified highly favorable areas, which accounted for approximately 60% and 42% of the study area according to Multi-Layer Perceptron and M5 Prime, respectively. These areas were predominantly located in the western region, characterized by mountainous terrain with relatively shorter slope lengths and altitudes ranging between 2500 m and 3500 m. This research serves as a valuable roadmap for decision-makers, offering guidance on how to improve the likelihood of discovering geomorphosites while minimizing costs and reducing the time required for exploration.

**Keywords** Geomorphosites · Machine learning · Geotourism · Geodiversity · Ziz · SE Morocco

## Introduction

Geotourism faces several significant challenges globally that threaten its sustainability and effectiveness. Environmental degradation, driven by increased tourist activity, poses a serious risk to fragile ecosystems and geological formations. Additionally, inadequate site identification often leads to the mismanagement of geosites, resulting in either overexploitation or neglect of valuable resources. Geosites are defined as geological sites that possess scientific, educational, or touristic significance. This category encompasses various features, including geological formations, fossils, minerals, and natural landscapes. Importantly, geosites include geomorphosites, which constitute a specific subset that emphasizes landforms and morphogenetic processes. Moreover, there is an urgent need for sustainable practices that harmonize tourism development with the preservation of natural and cultural heritage. Focusing on geoconservation, geoheritage, and the protection of geosites is crucial for sustainable development (Page 2024), as it promotes

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environmental stewardship and fosters community engagement in safeguarding our planet's geological treasures.

Recent research has provided valuable insights into addressing the challenges faced by geotourism. Chrobak et al. (2021) proposes innovative methodologies that tackle issues such as environmental degradation and site mismanagement, emphasizing the need for effective strategies in geotourism. Migoń et al. (2018) contribute to this discourse by enhancing our understanding of geosites, which is essential for the sustainable development of tourism in these areas. Furthermore, Rypl et al. (2021) illustrate the critical role of technology, particularly machine learning, in improving geotourism practices by enabling better site identification and management. Together, these studies underscore the importance of integrating innovative solutions and technological advancements to promote a sustainable future for geotourism.

Morocco is home to a rich geodiversity, featuring a diverse range of geological formations that have evolved over millions of years. These remarkable landscapes, encompassing magnificent mountains and breathtaking coastlines, are facing vulnerable circumstances. Nonetheless, the true potential of this geodiversity remains largely untapped, because it is inadequately explored, underutilized, and insufficiently promoted (Hamidy et al. 2024). Therefore, it is essential to prioritize the preservation of Morocco's geoheritage and implement effective geoconservation measures. By doing so, we protect the country's natural heritage, support scientific research, and promote sustainable tourism, ensuring a resilient legacy for future generations.

The growing interest in geoheritage for geotourism (Sen et al. 2024a, b), reflects a broader recognition of the importance of preserving our natural heritage while providing enriching experiences for travelers. So, geotourism, with its emphasis on sustainable exploration and interpretation of geodiversity, offers a unique opportunity to appreciate and understand the Earth's geological history (Anougmar et al. 2024). By fostering a deeper connection between visitors and the geological heritage of a region, geotourism not only promotes conservation but also contributes to local economies and community development.

According to the analysis of geoheritage studies conducted by Herrera-Franco et al. (2022), Italy (12.1%), Spain (8.77%), China (5.67%), Portugal (5.35%), and Brazil (5.31%) emerged as the top five countries in terms of scientific contributions in this field. In Morocco, El Hamidy et al. (2024) underscored, in their comprehensive overview of scientific research on geoheritage, the growing interest in this field. Furthermore, Elkaichi et al. (2024) demonstrated that there has been a rising recognition and emphasis on the concept of geoheritage at the national level, leading to an intensified focus on geoconservation initiatives. This has been accompanied by the emergence of geosite inventory initiatives undertaken by several Moroccan universities, notably the interesting study of the Jbel Ighoud site, regarded as the cradle of humanity (Hamidy et al. 2024; Baadi et al. 2023; Louz et al. 2023; Bussard 2023). These investigations frequently utilize techniques for inventorying and assessing with the objective of advancing and improving the conservation and appreciation of geoheritage (Louz et al. 2023).

In a recent study by ElKaichi et al. (2024), two popular geo-destinations in the Moroccan central High Atlas were examined and compared, further contributing to these efforts. These sites, with their geological significance and visual appeal, have become popular destinations for travelers seeking immersive and educational experiences.

However, it is worth noting that many of these studies primarily focus on sites of international significance, often overlooking regional and local sites that possess their own importance. In response to this, UNESCO, in collaboration with geopark territories, initiated the geopark network in 2004 as a means of promoting sustainable development in these territories (Elkaichi et al. 2023). Geoparks rely on two fundamental pillars to guide their strategy: leveraging geotourism to drive socio-economic growth and utilizing geoeducation to popularize geosciences. Consequently, the establishment of a geopark hinges upon the existence of geosites and geodiversity sites that possess significant scientific, tourism, and educational worth. Precise quantitative evaluations are employed to assess these values, enabling managers to propose appropriate geoconservation measures that contribute to territorial development.

However, creating an accurate inventory of geosites involves a systematic and comprehensive process to identify, document, and assess sites of geological significance. This includes surveying the region, conducting fieldwork, documenting geological features, and compiling the information into a database. The inventory process is widely regarded as the most efficient approach for mapping geosites (Nouhaila et al. 2024). However, it's important to recognize that inventory is just the initial step in gaining a comprehensive understanding of the potential heritage within a specific area.

Taking this viewpoint into consideration, it becomes essential to conduct a comprehensive inventory of sites that hold substantial scientific value, as well as geosites that offer educational and touristic appeal. These sites play a crucial role in disseminating the concepts of geoheritage and geoconservation to the local population.

The Ziz upper watershed (ZUW) encompasses a collection of geosites with significant heritage value, showcasing sedimentological, paleontological, karstic, and geomorphological features. The Faculty of Science and Technology of Errachidia can make effective use of these sites as both valuable training grounds and exciting excursion destinations. The region's geodiversity is derived from a wide range of geological formations, ranging from the Triassic era to the more recent Quaternary period (Manaouch et al. 2023a, b). Additionally, the area showcases a series of sedimentological, magmatic, and structural phenomena that have unfolded over time. Despite its remarkable richness, this geoheritage has seldom undergone a systematic inventory process and lacks adequate structures for promotion and conservation. Limited research has been conducted in this region, notably the study aimed at enhancing the geological heritage of the Errachidia area in the High Atlas, Morocco, which includes an inventory and a proposal for a pedagogical and geotouristic trail (Si Mhamdi et al. 2023). It presents breathtaking panoramic views along the edges of Wadi Ziz, stretching from Errich to Errachidia. The remarkable geomorphosites found at these edges have the potential to serve as geotourism stations. Inventorying geosites across the large, heterogeneous area of ZUW presents challenges, including limited resources, logistical constraints, and the region's diverse landscape. Accessibility issues and the vastness of the area further complicate the process.

To address this gap, the objective of this study is to inventory potential geomorphosites of the ZUW through a comprehensive assessment employing three popular machine learning algorithms. The evaluation outcomes will provide a solid basis for suggesting conservation and promotional initiatives that align with the guidelines and suggestions outlined by UNESCO and the Global Geoparks Network.

## Materials and Methods

### Study Area

The Draa-Tafilalet region, which contains the ZUW, spans approximately 4,435 km² in southeastern Morocco, located between latitudes 32° 5' 48" N and 32° 64' 19" N, and longitudes 4° 11' 72" W and 5° 46' 20" W (Fig. 1). Characterized by its elongated shape, the watershed is interspersed with numerous wadis that channel water toward the main Oued of Ziz, flowing eastward before turning southward.

Topographically, the ZUW exhibits a notable altitude range from 1,023 to 3,687 m above mean sea level, with certain areas exhibiting steep relief. The dominant slope class, comprising 80% of the region, falls within the 0–15° range (Mohamed et al. 2020).

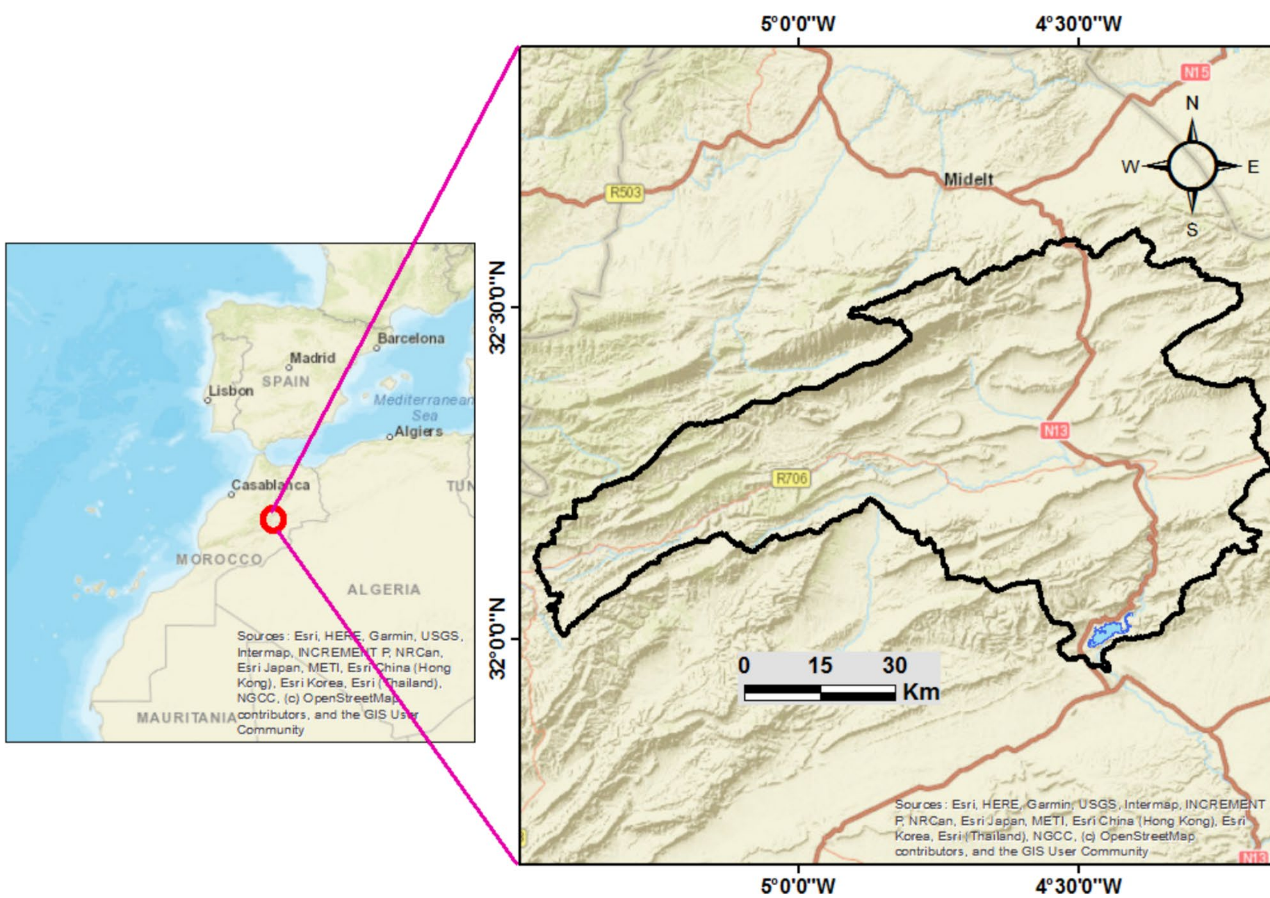



**Fig. 1** Location of Ziz upper watershed in South-East Morocco

The geological framework of ZUW, as illustrated in the geological map (Fig. 2), integrates data from two geological maps: Midelt at a scale of 1:200,000 and Rich-Boudnib at 1:200,000. The Triassic formations predominantly consist of detrital deposits, doleritic basalts, and occasionally evaporites, resting in angular unconformity atop a deformed Palaeozoic base shaped by multiple tectonic events (Michard 1976; Charrière 1990). The Jurassic series constitutes a significant portion of the Mesozoic strata within the High Atlas, to which ZUW belongs, and lies concordantly over the red formations of the Lower Triassic-Liassic. The lithology is primarily characterized by dolomites, limestones, marl-limestone alternations, and silico-clastic detrital deposits (Nouayti et al. 2015).

Structurally, the NW-SE extensional regime dominant during the Triassic-Jurassic periods dictated the formation of the Atlas basins, including ZUW. Rifting commenced in the Triassic, with the delineation and subsidence of these basins becoming evident during the Jurassic. The paleogeographic and morpho-structural landscape was characterized by tilted blocks primarily oriented NW or SE, bounded by significant normal faults trending NE-SW to ENE-WSW, interspersed with WNW-ESE transfer faults (Nouayti et al. 2015). In the Tertiary, a phase of biepisodic compression occurred, first in a NW-SE orientation, followed by N-S, which reactivated the older normal NE-SW faults into reverse faults or sinistral strike-slip thrusts. This compressive phase, which initiated the closure of the Atlas basins from the Upper Eocene, manifested during the Oligo-Miocene and Plio-Quaternary epochs (Michard 1976). ZUW encompasses numerous canyons and cliffs that exhibit the distinctive geological and geomorphological characteristics of the area. These sites offer valuable insights into the area's geological history and provide opportunities for scientific research, education, and tourism. One notable geomorphosite in ZUW is the Gorges of Ziz. These impressive limestone gorges have been carved by the Ziz River over millions of years, creating a dramatic landscape of steep cliffs and winding canyons. These geomorphological sites attract geologists and nature enthusiasts due to their geological significance and scenic beauty.

Precipitation in the Draa-Tafilalet region is generally infrequent and primarily occurs during autumn and spring. However, spatial variation in precipitation within ZUW is significant, attributed to the basin's topography: the western and northwestern regions are characterized by mountainous terrain with higher moisture levels, while the southern and eastern areas are markedly drier, influenced by Saharan conditions and the drying "Chergui" winds from the northeast in summer, alongside the "Sahéli" winds from the southwest in winter (Navas et al. 2009).

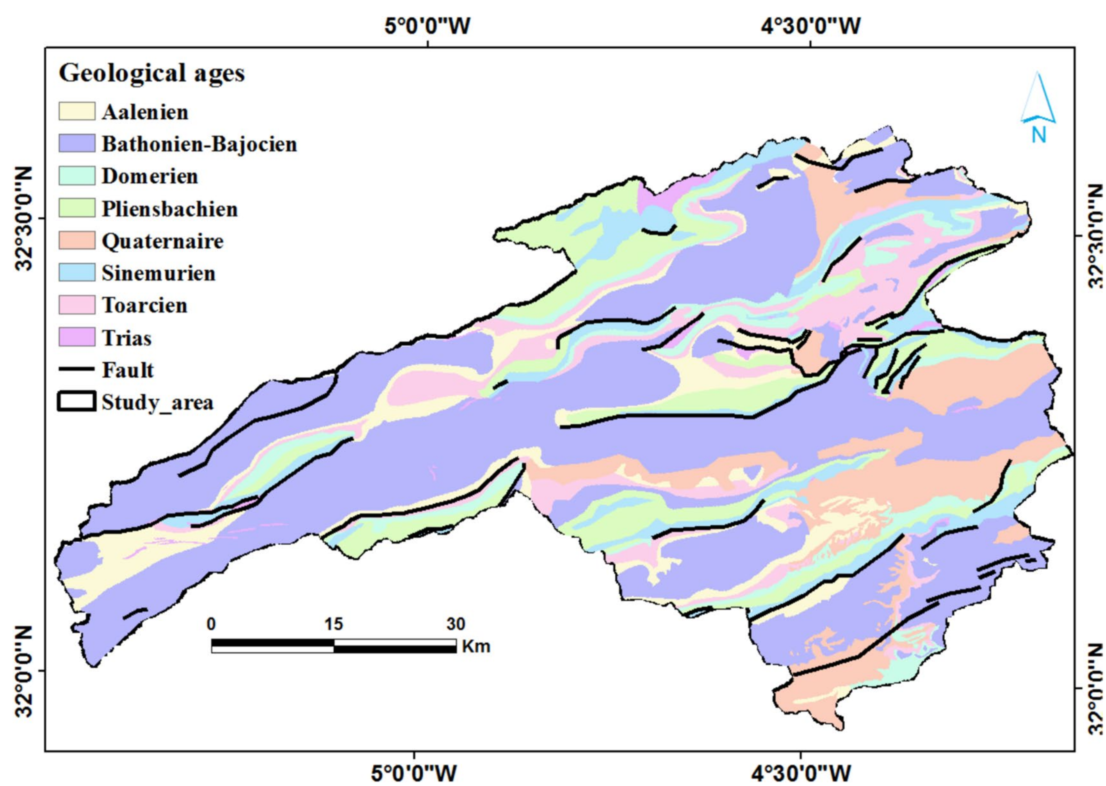



Fig. 2 Geology and structural setting of the study area.(Source: geological maps of Midelt at a scale of 1:200,000 and Rich-Boudnib at a scale of 1:200,000)

ZUW showcases a semi-arid climate marked by cold winters and dry summers (Manaouch et al. 2021). The annual average precipitation ranges from 119 to 377 mm, while the average yearly temperature varies between 10.2 and 19.2 °C (Manaouch et al. 2021).

The land use types identified and documented on the ZUW are as follows (Mohamed et al. 2020):

- Cultivated Fields: These are found in small parcels along watercourses.
- Water Bodies: Represented solely by the dam reservoir.
- Degraded Forest: This type is restricted to specific species, including Atlas cedar, Phoenician juniper, thuriferous juniper, Holm oak, Aleppo pine, and various thorny xerophytes.
- Poorly Vegetated areas: This category is predominantly characterized by extensive degradation and a significant scarcity of vegetation, attributed to ongoing irrational exploitation practices.

## Data used and Geomorphosites Inventory

In order to accurately pinpoint the areas where a specific phenomenon will occur necessitates a thorough analysis of the interplay between the location where the phenomenon has manifested and the underlying factors that have contributed to its occurrence (Al-Abadi 2018). Therefore, to ensure the accurate selection of geomorphosite sites, multiple field visits were conducted in ZUW. A total of 120 geomorphological sites were identified in the area, alongside the consideration of an additional 120 non-geomorphosite locations (Fig. 3). The methodology of considering 120 geomorphosites and 120 additional non-geomorphosite sites will enable a comparison of these 240 sites to evaluate, refine, and understand machine learning models, thereby enhancing the accuracy and reliability of predictions.

Topographic data, such as elevation, slope, and landform classification, helps in identifying and assessing the geomorphological features and processes that make a site significant from a geological or geomorphological perspective. The present study incorporated three conditioning factors, including topographic data, to assess geomorphosites. Altitude, slope length, and slope were considered as independent variables in determining the suitability of areas for locating geomorphosites. These variables were measured for the 240 selected sites in the inventory, as depicted in Fig. 3. To analyze the data, the dataset was divided into two distinct categories: training data (70%) and validation data (30%), a commonly utilized methodology in geospatial modeling (Manaouch et al. 2023a, b).

The data processing was conducted using ArcGIS 10.5 and SPSS Statistics 26 software. These software tools

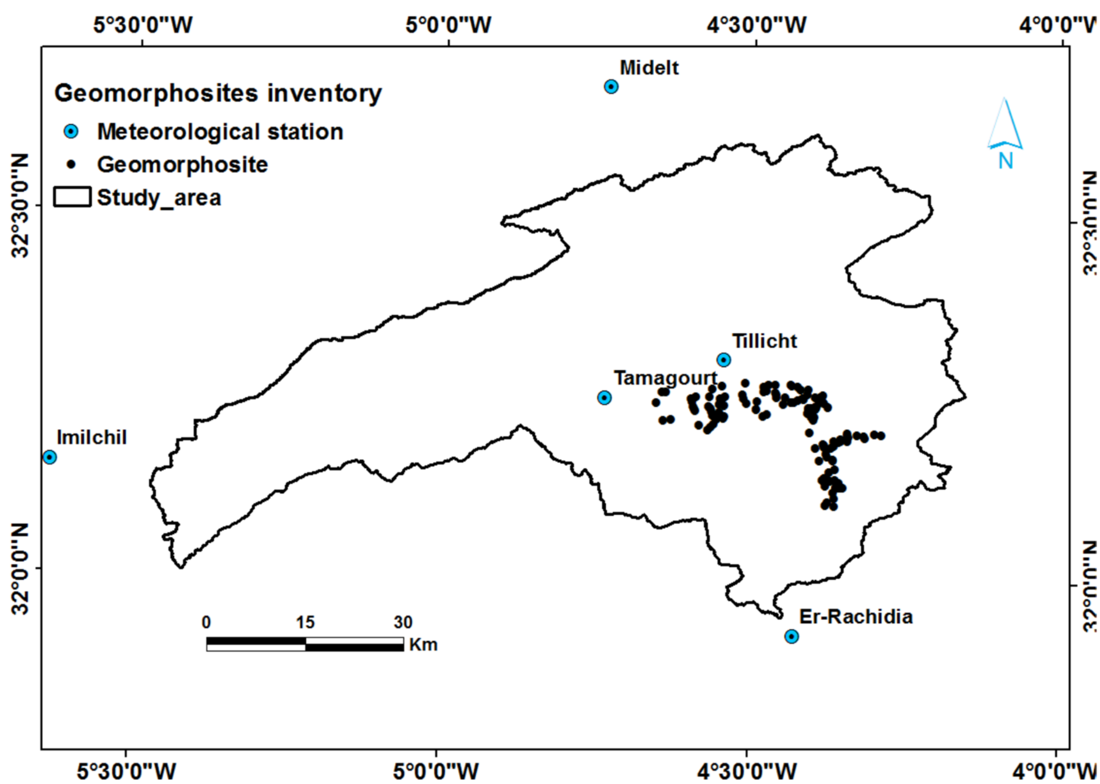



Fig. 3 Geomorphosites inventory map

possess geoprocessing capabilities, which facilitate the conversion of geospatial data into a tabular format. This conversion involves multiple steps, including importing geospatial data, defining attributes, running geoprocessing tools, configuring geoprocessing parameters, and executing geoprocessing operations.

### Geomorphological Sites: Mountains, Hills, Valleys, Plateaus, Canyons, Cliffs, Lakes and Rivers

ZUW demonstrates significant geodiversity, making it particularly attractive for geotourism. Multiple field visits have documented a diverse range of landforms, including canyons, lakes, caves, valleys, plateaus, and cliffs. Canyons and cliffs serve as essential components of geotourism, as these locations are recognized for their geological significance and offer opportunities for scientific, educational, cultural, and aesthetic engagement. They showcase unique geological features such as rock formations, fossils, geomorphological landscapes, or remarkable natural phenomena. These sites are often prime locations for the study of geology and provide opportunities for learning and raising awareness about the Earth sciences.

Cliffs are vertical geological formations composed of hard rocks that tower above the landscape. They can result from processes such as erosion, plate tectonics, or sedimentation. Cliffs often provide spectacular geological exposures, allowing scientists and visitors to visualize geological layers and study the geological history of a region.

Canyons are defined as deep, narrow valleys with steep sides, typically formed by the erosive action of rivers or other natural forces over an extended period. Canyons often exhibit unique geological features, such as exposed rock layers, intricate formations, and impressive landscapes, making them significant and visually captivating geomorphosites.

Canyons and cliffs in the ZUW region are vital for geotourism, providing opportunities for education, exploration, and the discovery of geological wonders. These natural features not only enhance the experience of visitors but also contribute to the sustainable economic growth of local communities. Figure 4 presents visual representations of various canyons, cliffs, and guided tours located within the study area, highlighting their significant geological features.

## Methodology

Figure 5 presents a flowchart illustrating the process of mapping potential geomorphosites using three popular machine learning algorithms. Further elaboration on the details of this approach is provided below.

### Geomorphosites Conditioning Factors (GCFs)

#### Slope

The slope is particularly relevant to the objectives of this study for several reasons. First, it influences the accessibility of a site. A slope that is too steep can make access difficult, or even dangerous, for visitors, which may discourage geotourism. Additionally, the slope plays a crucial role in the safety of outdoor activities. Steep terrains can present risks of accidents, such as falls or slips, raising concerns for geotourists. Furthermore, the slope can affect the geological stability of a site. Unstable slopes can lead to landslides, making a location less suitable for prolonged visits. Therefore, understanding the slope helps assess the viability of a site for geotourism, considering both accessibility and safety. This aids in identifying the most appropriate locations for developing geotourism activities while ensuring the safety and comfort of visitors.

In geotourism, slope (Fig. 6) plays a vital role in shaping landscapes and influencing visitors' experience. Steep slopes, such as those found in canyons and cliffs, often create spectacular landscapes that attract tourists and enhance recreational opportunities such as hiking and photography. As a result, several studies have presented typical geosites of canyons formed due to the tectonic activity, break of slope forming a seasonal waterfall (Sen et al. 2024a, b), sea cliff and sea caves (Jon et al. 2024), water bodies formed in the base of the canyon (locally known as Jheels) (Baitalik 2024). The slope of a geosite is a crucial characteristic that influences its appearance, stability, and geological processes. Geologists and geographers use various methods to measure and describe slopes, considering the angle of inclination and the rate of change in elevation. Alternatively, the slope (m) can be determined using the coordinates of two points (x, y) and (x', y') on the line:

$$m = \left(y - y'\right) / \left(x - x'\right) \tag{1}$$

The slope can be positive, negative, zero, or undefined, depending on the direction and nature of the line or curve. A positive slope signifies an upward trend, while a negative slope indicates a downward trend. A slope of zero represents a horizontal line, and an undefined slope is associated with a vertical line.

#### Length of Slope

Length of slope plays a crucial role in determining geomorphosite potential by influencing erosion rates, landform stability, and visitor attraction. Longer slopes typically experience higher erosion rates due to increased


**Fig. 4** Examples of the geological features in the studied area: (**a**) Jurassic tabular structures, (**b**) wadi Ziz canyon, (**c**) Guided tours for tourists to the thermal spring “Hammat Moulay Ali Cherif”, (**d**) Hassan Addakhil Dam Lake, (**e**) Tunnel “Zaabel”, (**f**) Ziz Valley “Ait Atmane”, (**g**) The waterfall and (**h**) Ziz Gorges. (Source: Jbdodane https://www.flickr.com/photos/jbdodane)

runoff and sediment transport, which can lead to more pronounced landforms and features. Conversely, shorter slopes may promote stability, allowing for the preservation of unique geological formations. Additionally, the scale of visible landforms associated with varying slope lengths can enhance their attractiveness to visitors, as dramatic landscapes often draw more interest. Thus, understanding slope length is essential for assessing both the ecological processes at play and the potential for tourism in a given area. Therefore, the length of slope (Fig. 7) is an important parameter used to describe and characterize landforms. Measuring the length of slope helps in quantifying the extent and scale of these landforms. It can be determined by measuring the horizontal distance along the slope’s contour or by using topographic maps, aerial imagery, or satellite data. Field surveys and remote sensing techniques are commonly employed to obtain accurate measurements. It provides information about the size and dimensions of the landform, aids in assessing the potential for geomorphological processes such as erosion or mass wasting and contributes to the characterization of the geosite’s overall geomorphological context. If the angle of the slope and the vertical elevation change are known, trigonometry can be used to calculate the length of the slope. The formula used is:

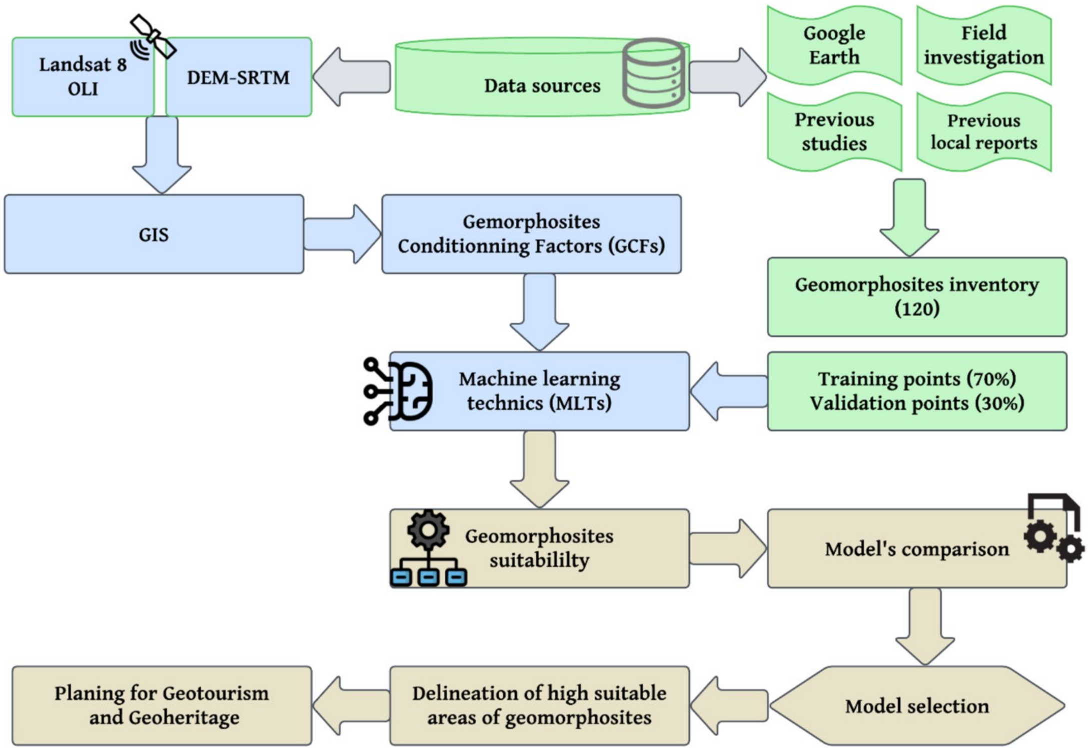
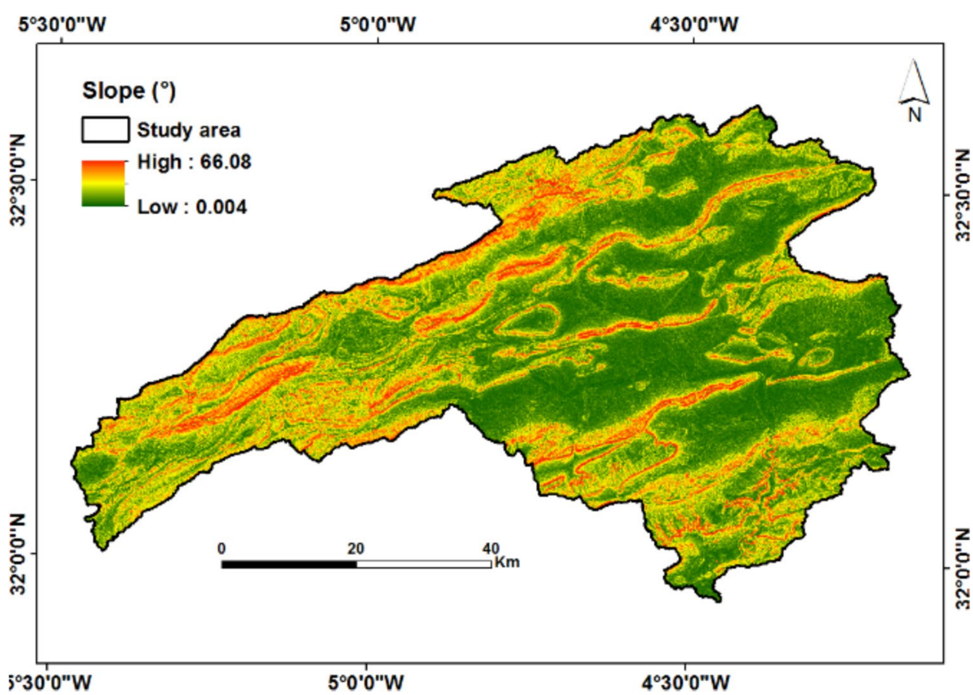



**Fig. 5** Flowchart of the methodology used in the study

**Fig. 6** Geomorphosites conditioning factor (GCF) used in this study: Slope



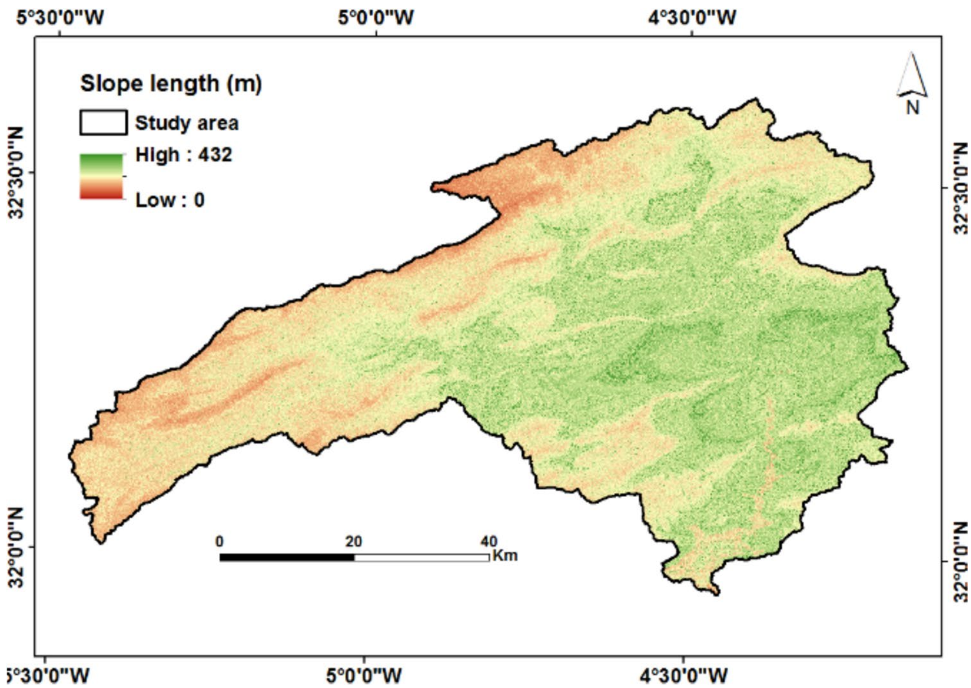



**Fig. 7** Geomorphosites conditioning factor (GCF) used in this study: Slope length

$$Length\ of\ slope = Vertical\ elevation\ change\ /\ \sin\ (angle\ of\ slope) \tag{2}$$

## Elevation

Elevation significantly enhances geomorphosite attractiveness by providing panoramic views that can captivate visitors, making higher sites particularly appealing for geotourism. Additionally, elevated areas often feature unique environmental conditions, which can enrich the visitor experience. Accessibility is another factor; while some high-elevation sites may be challenging to reach, those that are easily accessible can attract more visitors. These elements align with geomorphosite selection criteria, emphasizing aesthetic value, ecological significance, and visitor engagement, thereby highlighting the importance of elevation in promoting geomorphosite potential. Therefore, elevation (Fig. 8) is an important characteristic used to describe and differentiate various landforms. Measuring elevation is typically done using surveying techniques, satellite-based systems (e.g., GPS), or remote sensing methods such as LiDAR (Light Detection and Ranging) or radar. These technologies provide accurate elevation data that can be used to create digital elevation models (DEMs) or topographic maps. In the context of geomorphosites, elevation pertains to the vertical distance or height of a specific point or location above a reference datum. It is a fundamental parameter used to describe the vertical position of landforms and is essential for understanding the characteristics, processes, and geological significance of geosites.

## Machine Learning Algorithms (MLA) and Frequency Ratio (FR)

Machine learning classifiers (MLCs) have become increasingly prominent in addressing spatial modeling challenges within the realm of geotourism, particularly in the context of tasks such as monitoring landscape changes in geoparks (Pham et al. 2024). The subsequent paragraphs provide a detailed account of the machine learning classifiers employed within the scope of this research.

### M5 Prime (M5P)

The M5P algorithm, also known as the M5 Prime algorithm, is a machine learning algorithm used for generating decision trees. It is primarily designed for regression tasks and is particularly effective when dealing with datasets where most of the attributes are numeric.

The M5P algorithm, an extension of M5, consists of four stages. It splits the input vector into sub-spaces by selecting attributes and split points that optimize data separation. This process creates branches and nodes in the decision tree, capturing complex attribute-target relationships. (Saha et al. 2024). This algorithm has been utilized in numerous works spanning multiple disciplinary domains such hydrological phenomena (Asadi et al. 2024), estimating soil water content (Emami et al. 2024), assessment of groundwater potential (Sharma et al. 2024).

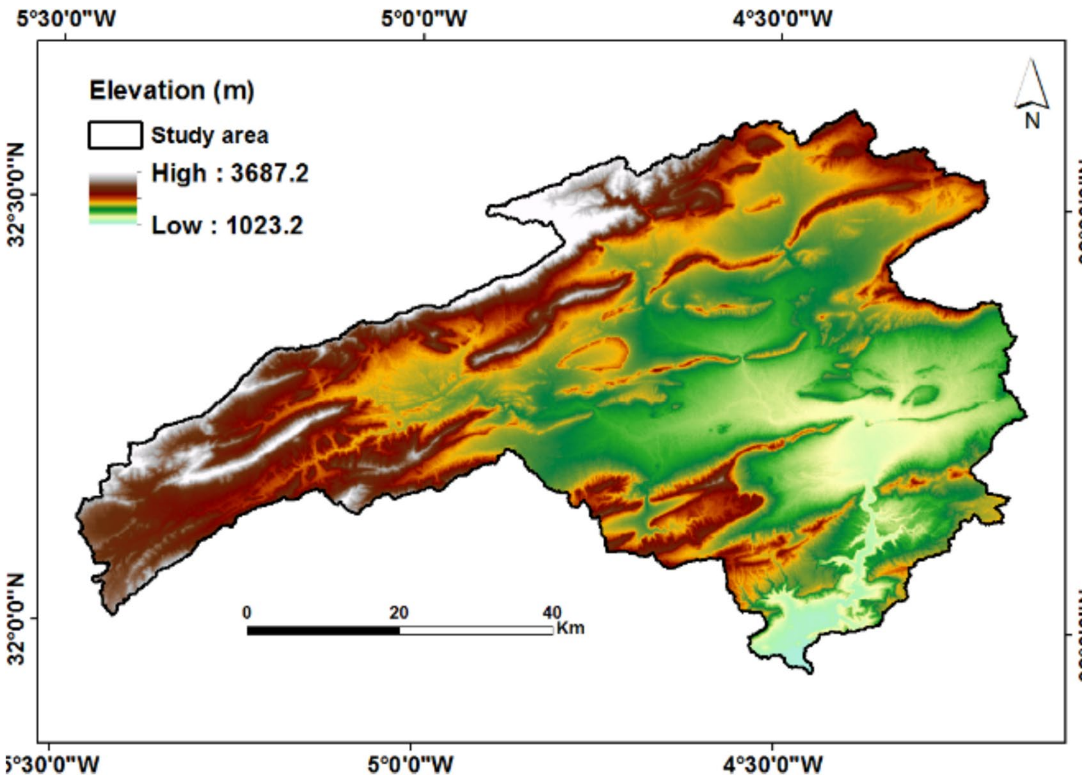



**Fig. 8** Geomorphosites conditioning factor (GCF) used in this study: Elevation

#### Multi-Layer Perceptron (MLP)

The Multi-Layer Perceptron (MLP) is an artificial neural network (ANN) architecture comprising multiple interconnected layers of nodes, also referred to as artificial neurons. It operates as a feedforward neural network, wherein information propagates in a unidirectional manner, flowing from the input layer through the hidden layers and ultimately reaching the output layer.

The MLP supports both classification and regression tasks. The MLP algorithm learns by adjusting the weights of the connections between nodes during the training process. The key components and parameters of the MLP are five: Number of hidden layers; Number of nodes in each hidden layer; Learning rate; Momentum and Training iterations.

MLP has been evaluated as a valuable and versatile tool across various geospatial modeling applications, including the analysis of drought patterns (Mansour 2024), in the prediction of potential reforestation areas (Manaouch et al. 2023a, b), and in tourism landscape planning (Siroosi et al. 2020).

#### Random Forest (RF)

Random Forest is a widely adopted ensemble learning algorithm that is utilized for both classification and regression tasks. It leverages the principles of decision trees and integrates the predictions generated by numerous individual decision trees to produce final predictions. The key aspects and parameters of Random Forest include Number of trees; Random feature selection: Out-of-bag (OOB) error and Feature importance.

Random Forest (RF) is a highly utilized algorithm in geospatial modeling, finding applications in various domains such as predicting land use changes (Ouma et al. 2024), assessing landslide susceptibility (Manaouch et al. 2023a, b), and many more. Its versatility and effectiveness make it a popular choice in the field of geospatial analysis.

#### Frequency Ratio (FR)

According to Yousefi et al. (2021), the frequency ratio (FR) is emphasized as a commonly employed statistical technique for examining the spatial correlation between locations and categorical factors. FR calculates values and success percentages of each class based on the presence or absence of geomorphosites using the following Eq. (1):

$$FR = {}^{\left(L_i K_{ji} * \sum pix\right)} \Big/ {}_{\left(L_i K_{ji} * \sum Li\right)} \quad (3)$$

**Table 1** Spatial relationship between topographical factors and geomorphosites using frequency ratio

| GCF | Class | No. of pixels in class | % of class | No. of sites | % of sites | Frequency Ratio |
|---|---|---|---|---|---|---|
| Slope (°) | 1 (< 15) | 1809027 | 36,7839 | 2 | 0,01 | 0,04 |
| | 2 (15–30) | 1250863 | 25,4344 | 9 | 0,07 | 0,3 |
| | 3 (30–45) | 881995 | 17,9340 | 16 | 0,13 | 0,74 |
| | 4 (45–60) | 672053 | 13,6652 | 24 | 0,2 | 1,42 |
| | 5 (> 60) | 304042 | 6,18225 | 69 | 0,57 | 9,58 |
| Elévation (m) | 1 (< 1399) | 393526 | 8,00 | 1 | 0,008 | 0,10 |
| | 2(1399–1587) | 850896 | 17,30 | 4 | 0,03 | 0,19 |
| | 3(1587–1775) | 837542 | 17,03 | 14 | 0,11 | 0,68 |
| | 4(1775–1963) | 885384 | 18,00 | 0 | 0 | 0 |
| | 5(1963–2161) | 655483 | 13,33 | 3 | 0,02 | 0,19 |
| | 6(2161–2381) | 552065 | 11,23 | 0 | 0 | 0 |
| | 7(2381–2621) | 409303 | 8,32 | 73 | 0,60 | 7,60 |
| | 8(2621–2935) | 240897 | 4,90 | 25 | 0,20 | 4,16 |
| | 9(> 2935) | 92885 | 1,89 | 0 | 0 | 0 |
| Length of Slope (m) | 1(< 59) | 21828 | 0,44371 | 0 | 0 | 0 |
| | 2(59–161) | 76772 | 1,56058 | 0 | 0 | 0 |
| | 3(161–208) | 1015571 | 20,6441 | 4 | 0,03 | 0,15 |
| | 4(208–264) | 1949329 | 39,6251 | 24 | 0,2 | 0,5 |
| | 5(> 264) | 1855923 | 37,7264 | 92 | 0,76 | 2,01 |

**Table 2** Spatial relationship between topographical factors and non-geomorphosites using frequency ratio

| GCF | Class | No. of pixels in class | % of class | No. of sites | % of sites | Frequency Ratio |
|---|---|---|---|---|---|---|
| Slope (°) | 1 (< 15) | 1809027 | 36,7839 | 86 | 0,71 | 1,93 |
| | 2 (15–30) | 1250863 | 25,4345 | 23 | 0,19 | 0,76 |
| | 3 (30–45) | 881995 | 17,9341 | 9 | 0,075 | 0,41 |
| | 4 (45–60) | 672053 | 13,6652 | 1 | 0,008 | 0,05 |
| | 5 (> 60) | 304042 | 6,1823 | 1 | 0,008 | 0,13 |
| Elévation (m) | 1 (< 1399) | 393526 | 8,0018 | 0 | 0 | 0 |
| | 2(1399–1587) | 850896 | 17,3017 | 0 | 0 | 0 |
| | 3(1587–1775) | 837542 | 17,0302 | 0 | 0 | 0 |
| | 4(1775–1963) | 885384 | 18,0030 | 0 | 0 | 0 |
| | 5(1963–2161) | 655483 | 13,3283 | 5 | 0,041 | 0,32 |
| | 6(2161–2381) | 552065 | 11,2254 | 0 | 0 | 0 |
| | 7(2381–2621) | 409303 | 8,3226 | 35 | 0,29 | 3,64 |
| | 8(2621–2935) | 240897 | 4,8983 | 80 | 0,66 | 13,33 |
| | 9(> 2935) | 92885 | 1,8887 | 0 | 0 | 0 |
| Length of slope (m) | 1(< 59) | 21828 | 0,4437 | 0 | 0 | 0 |
| | 2(59–161) | 76772 | 1,5606 | 0 | 0 | 0 |
| | 3(161–208) | 1015571 | 20,6441 | 3 | 0,025 | 0,11 |
| | 4(208–264) | 1949329 | 39,6252 | 14 | 0,11 | 0,29 |
| | 5(> 264) | 1855923 | 37,7264 | 103 | 0,85 | 2,25 |

Li represents the count of sites within each factor class Kji. Σpix denotes the total number of pixels in the factor map of ZUW, while li represents the total count of pixels in each factor class Kji. Additionally, ΣLi represents the overall count of geomorphosites.

The frequency ratio (FR) values were calculated for the three conditioning factors, each comprising multiple classes. These FR values have been tabulated and are presented in both Tables 1 and 2. The spatial relationship between GCFs and geomorphosites was assessed by calculating the FR, and

**Table 3** Data labeling (predicted and actual)

| Actual label | Predicted label | |
|---|---|---|
| | Positive | Negative |
| Positive | True Positive (TP) | True Negative (TN) |
| Negative | False Positive (FP) | False Negative (FN) |

**Table 4** Correlation matrix

| | Slope | Length of slope | Elevation |
|---|---|---|---|
| Slope | 1 | −0.17 | −0.32 |
| Length of slope | −0.17 | 1 | 0.15 |
| Elevation | −0.32 | 0.15 | 1 |

the results for all factors are shown in the following tables (Tables 1 and 2).

To ascertain the spatial correlation between GCFs and non-geomorphosite points, the frequency ratio (FR) was computed. The results of the FR analysis for the three factors are presented in Table 2.

### Performance and Model's Validation

In any modeling process, it is crucial to prioritize result validation. Geospatial modeling studies commonly employ the area under the curve of the receiver operating characteristics (AUC/ROC) method, as highlighted by Manaouch et al. (2021), to evaluate and validate the accuracy and performance of the models. In this modeling study, the AUC/ROC analysis involved the assessment of 120 geomorphosite sites using classified maps that depicted the suitability of geomorphosites within the study area. The geomorphosite suitability maps were categorized into four groups: true positive (TP), true negative (TN), false positive (FP), and false negative (FN) events, as detailed in Table 3.

Afterwards, the true positive rate (TPR) and false positive rate (FPR) were computed using Eqs. (2) and (3) respectively. To visualize the performance of the classifier, an ROC curve was constructed by plotting the FPR on the X-axis and the TPR on the Y-axis:

$$TPR = TP/(TP + FN) \tag{4}$$

$$FPR = FP/(FP + TN) \tag{5}$$

Based on the evaluation criteria for the area under the curve (AUC), different classes are defined as follows: AUC values of 0.5–0.6 are considered weak, 0.6–0.7 as moderate, 0.7–0.8 as good, 0.8–0.9 as very good, and values above 0.9 as excellent (Yousefi et al. 2021).

## Results

### Correlation between GCFs

To ensure objectivity and independence of the geomorphosite conditioning factors (GCFs), a statistical analysis method using Pearson's correlation coefficient was employed for the 3 preselected factors. The results of the correlation analysis are presented in Table 4, indicating that all the conditioning factors satisfy the independence criteria.

### Analysis of Geomorphosites Conditioning Factors

To evaluate the significance of Geomorphosites Conditioning Factors (GCFs), the "CorrelationAttributeEval" attribute evaluator was employed. This evaluator was used to assess the predictive capability of the three GCFs. The relative importance of each factor is depicted as a percentage on the X-axis in Fig. 9. The results obtained from the CorrelationAttributeEval indicated that the slope exhibited the highest contribution (58%), followed by elevation (50%) and length of slope (ls) (21%) appeared to have relatively less importance in the context of potential geomorphosites modeling. The findings of the study conducted by Zakharovskyi et al. (2024) provide further confirmation that the slope plays a vital role in determining geosites. Their approach involves employing an 8-point evaluation system to assess the geological formations within the study area, which is then multiplied by the degree of morphological slope. They indicate that the slope model serves as one of the most effective data sources for the qualitative-quantitative assessment of geodiversity, functioning as a proxy for the classification of geomorphological sites. Their approach involves employing an 8-point evaluation system to assess the geological formations within the study area, which is then multiplied by the degree of morphological slope. These results complement and reinforce our own findings related to slope dynamics.

### Geomorphosites and Conditioning Factors

The initial assessment involved calculating the frequency ratio (FR) values to explore the relationships between the conditioning factors and geomorphosites. The results indicated that certain classes or categories, such as altitude classes 4, 6, and 9, and length of slope classes 1 and 2, had an FR value of 0 (Table 1). However, it is important to note that the presence or frequency of sampled sites (120 sites) in each class does not necessarily imply a higher favorability for geomorphosites compared to other classes.

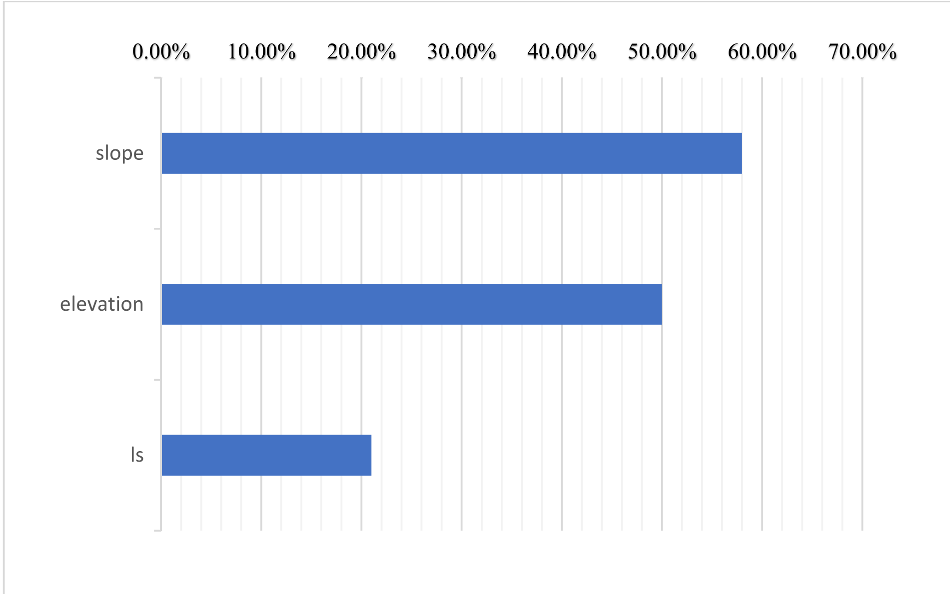



**Fig. 9** GCFs and the corresponding predication rate by average merit

Conversely, categories or classes exhibiting higher FR values are regarded as favorable or suitable areas for geomorphosites. These include slope class 5, altitude class 7, and length of slope class 5. These factors exhibited substantial FR values, indicating their significance in supporting the presence of geomorphosites.

## Geomorphosites Suitability Map

To facilitate the comparative analysis, Figs. 10 , 11, and 12 present the three geomorphosites suitability maps generated by the M5P, MLP and RF classifiers. For each classifier, four suitability classes (Low, Medium, High, and Very high)

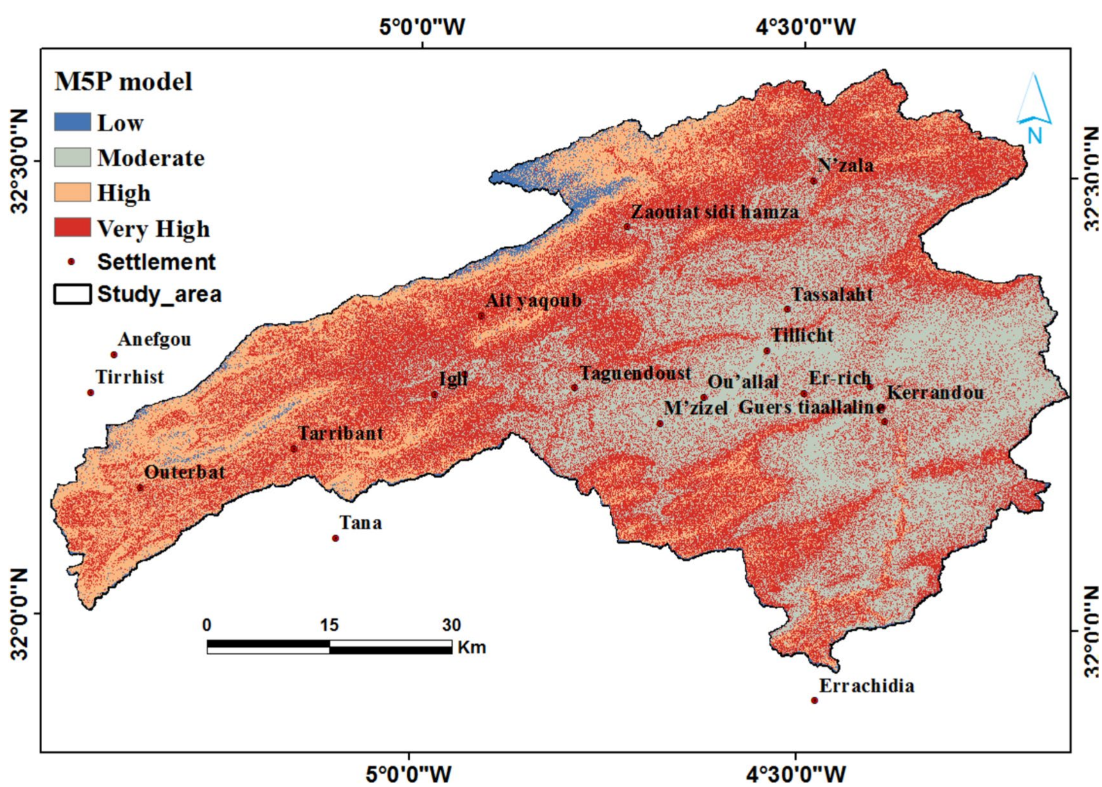



**Fig. 10** Generated geomorphosites suitability map using M5P model

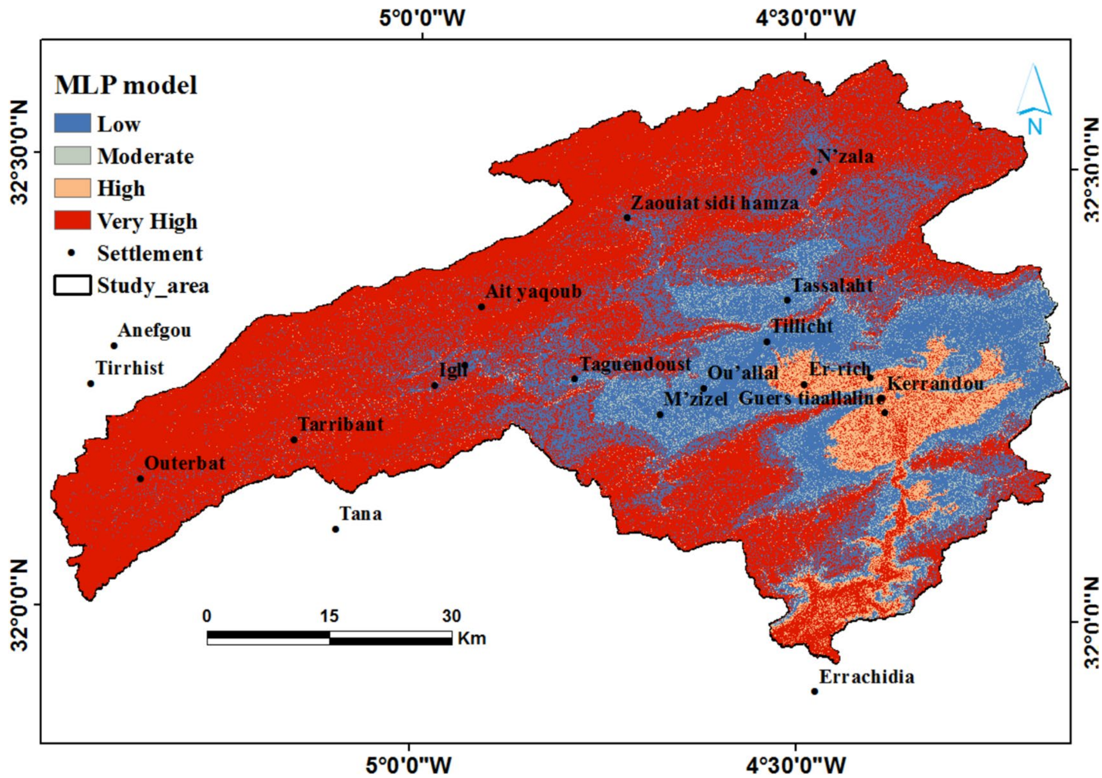



**Fig. 11** Generated geomorphosites suitability map using MLP model

were assigned to assess geomorphosites suitability. To shed light on the reasons behind the disparities and to understand why certain areas are deemed suitable while others are not, we have sought the expertise of individuals with extensive knowledge of the region. In addition, we have cross-referenced their insights with the images provided by Google Earth to validate our findings. The respective area percentages of these classes were subsequently calculated for each model, as depicted in Fig. 14.

### Performance Evaluation and Comparative Analysis

To assess the effectiveness of the classifiers, the AUC/ROC method was utilized. During the training phase, the obtained results indicated that MLP (Multi-Layer Perceptron), M5P, and RF (Random Forest) achieved average accuracy rates of 100%, 97.29%, and 94.24% respectively.

During the validation phase, notable variations in classifier performance were observed (Fig. 13). The AUC values ranged from 0.687 to 0.911, with MLP attaining the highest value of 0.911, followed by M5P with 0.779, and RF with 0.671. Both M5P and MLP classifiers exhibited AUC values exceeding 0.75, indicating their effectiveness in modeling geomorphosites suitability mapping in ZUW. However, M5P demonstrated higher reliability and accuracy compared to MLP and RF, making it the more robust choice for this task.

## Discussion

### Analysis of GCFs

The analysis of conditioning factors plays a crucial role in modeling studies as it enables the identification of key parameters that contribute to the occurrence of events. In the context of geomorphosites modeling, this study identified three factors as significant.

When considering geomorphosites, the slope gradient emerges as the most influential factor due to its impact on a range of geomorphic processes and landform development. Steep slopes frequently reveal remarkable landforms, such as badlands, which refer to rugged landscapes characterized by steep slopes, deep ravines, and minimal vegetation, typically formed by the rapid erosion of soft, unconsolidated materials. These "badlands" are highly valued for their aesthetic and scientific importance in the field of geomorphology (Kumar et al. 2024). The inclusion of steep slopes enhances the visual appeal and scenic allure of a geomorphosite.

Altitude is a critical variable in the evaluation of geomorphosites, as it significantly affects their formation and characteristics. In high-altitude regions, geomorphosites often exhibit distinct features that differentiate them from those found in lower-altitude areas, which may present alternative relief forms and geological attributes that may not qualify

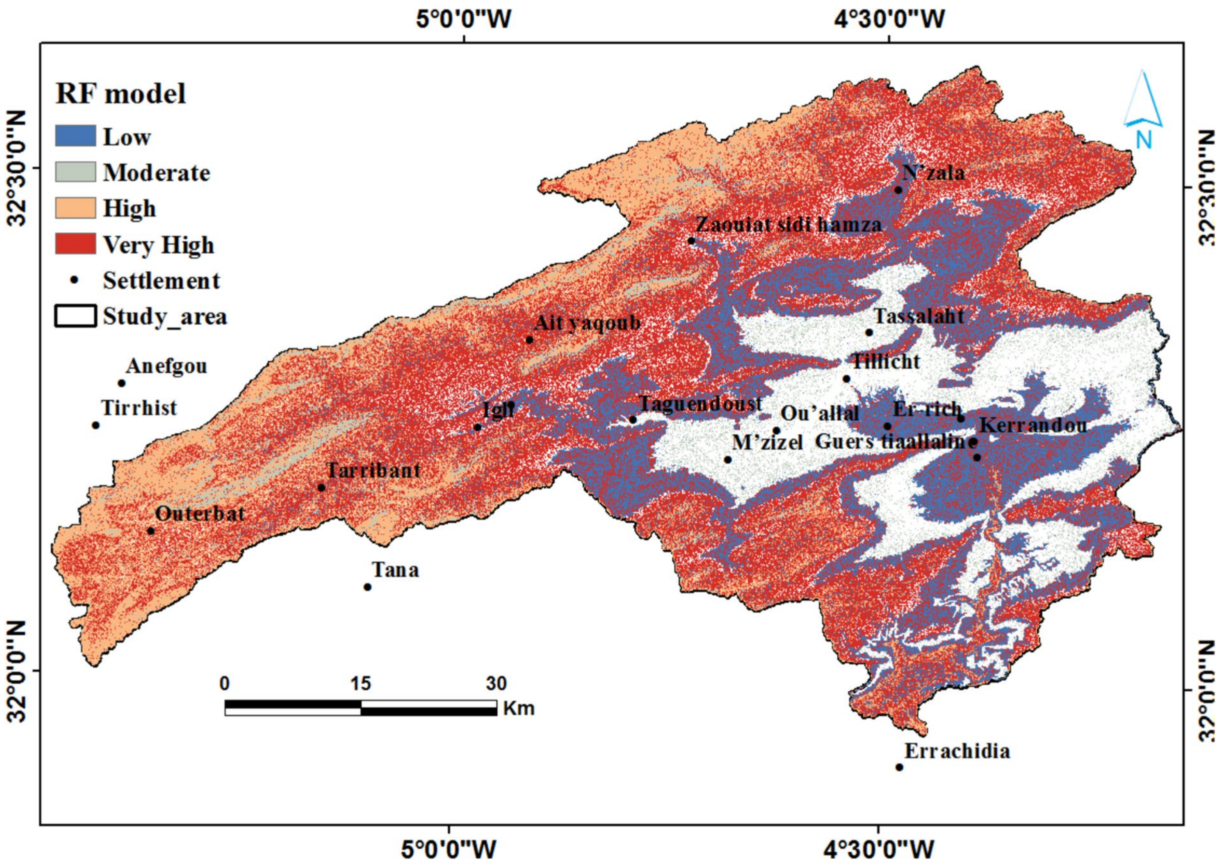



**Fig. 12** Generated geomorphosites suitability map using RF model

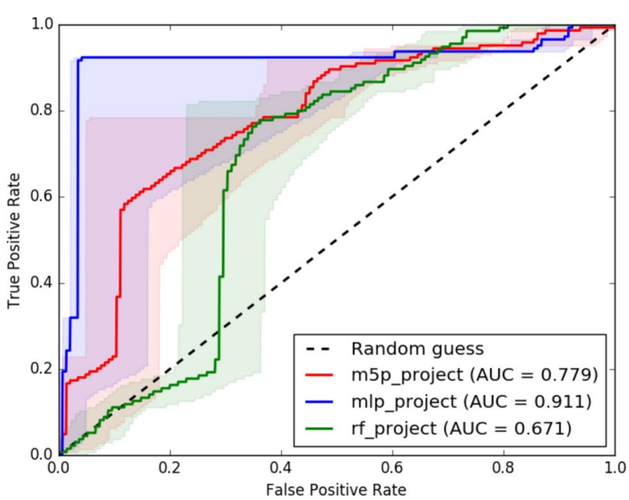



**Fig. 13** Prediction rate curves for the suitability maps of geomorphosites

as geomorphosites. For instance, in the Upper Ziz region, the mountainous terrain to the west displays considerable geodiversity, aligning with Guerini et al. (2024) findings that highlight the complexities inherent in such geologically diverse environments. These complexities can pose challenges for geomorphological assessment and conservation efforts.

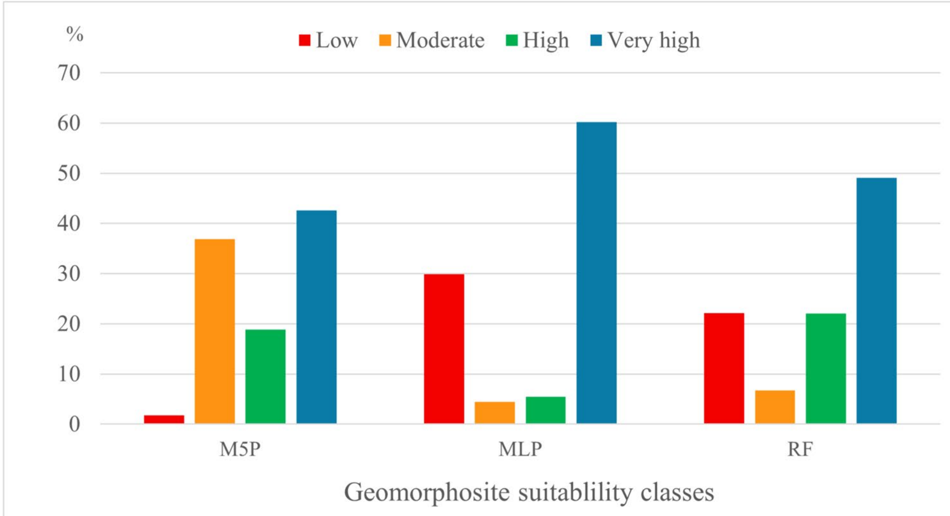
Furthermore, the analysis revealed that the length of slope factor contributed in descending order of importance. Although still relevant, its impact was relatively lower compared to the slope and elevation factors.

In conclusion, this analysis provides valuable insights into the conditioning factors that specifically influence the occurrence of geomorphosites in the ZUW region.

## Classifier Building

The selection of machine learning classifiers plays a crucial role in accurately mapping potential geomorphosites. For making comparisons, it is important to note that no prior research is directly comparable to the current study, as it uniquely concentrates on the ZUW while integrating specific factors related to the prediction of geomorphosite potential, thereby highlighting its innovative contribution to the field. In this study, three popular classifiers, namely Random Forest (RF), Multilayer Perceptron (MLP), and

Fig. 14 Percentage of Geomorphosite Suitability Classes' areas for three applied MLCs




M5P, were tested due to their favorable results in previous modeling studies.

RF is recognized for its advantageous properties, including fewer restrictions on variables, disregard for data scale and distribution, high computational efficiency, and high accuracy. However, in this study, the performance of the RF classifier was limited to a maximum accuracy of 68%, which is notably lower than the 88.3% accuracy achieved in the identification of potential archaeological sites in central Jharkhand, India (Pal et al. 2024). On the other hand, MLP, which consists of layers and neurons and employs backpropagation algorithms, demonstrated the highest performance in this study, aligning with a realistic simulation of drought occurrence in Oman (Mansour 2024). In this study, the performance of the M5P classifier, which is less frequently utilized in geospatial modeling, surpassed 77%. However, it was noted that a higher accuracy of 92.5% was achieved in landslide susceptibility modeling, as reported by Saha et al. (2024).

Overall, the choice of classifier significantly influences the accuracy of mapping geomorphosites' suitability. Despite its strong performance in previous studies, the RF classifier did not achieve the same level of accuracy in this study. These RF results could be attributed to several factors. Firstly, the characteristics of the data used may not align well with the model's assumptions, potentially leading to suboptimal performance. Additionally, regional specificities, such as unique environmental conditions or variations in geomorphological features, might have influenced the outcomes, affecting the model's ability to accurately predict geomorphosite potential. MLP exhibited the highest performance, consistent with other studies, while M5P, despite being less commonly used, showed promising accuracy.

### Analysis of Precision Comparisons

The performance of machine learning classifiers (MLCs) varies depending on the chosen algorithm. Through the utilization of the 10-fold cross-validation method during the training phase, RF, M5P, and MLP demonstrated varying levels of performance. However, during the validation phase, their performance decreased to 67.1%, 77.9%, and 91.1%, respectively.

In numerous studies, model validation is conducted through the calculation of the AUC-ROC. For instance, Roy et al. (2024) evaluated potential suitability mapping for ecotourism development in Darjeeling, reporting an AUC-ROC value of 0.845, which indicates a high level of reliability in relation to the actual conditions. Additionally, Manaouch et al. (2023a, b) utilized the AUC-ROC method to compare results in their analysis of potential reforestation areas. It is generally observed that algorithms with higher AUC values demonstrate more accurate and efficient prediction capabilities (Su et al. 2021). Analyzing the ROC curves of the three classifiers, it can be observed that they exhibit high performance during the training phase. However, in the validation phase, MLP demonstrated the highest AUC value of 0.911, followed by M5P with 0.779 and RF with 0.671. Considering the AUC values of the three models, MLP outperforms the others.

### Distribution of Potential Geomorphosites Areas

The attribute data from ZUW was preprocessed using the stored versions of three classifiers: RF, MLP, and M5P. This preprocessing included the computation of geomorphosites suitability indices for each pixel within the study area. To classify the geomorphosite suitability indices, the "Natural breaks" algorithm was employed, resulting in the creation of four classes: low, moderate, high, and very high. As a result, three distinct geomorphosite suitability maps were generated, as illustrated in Figs. 10 , 11, and 12.

The findings reveal that areas classified as very highly suitable for potential geomorphosites account for 42.57%, 60.2%, and 49.07% of the entire study area for M5P, MLP, and RF respectively (Fig. 14). All classifiers indicate that south parts exhibit low to moderate suitability of potential geomorphosites due to their flat terrain and generally low elevations. These features restrict the development of significant geomorphological processes, leading to less favorable conditions for ecological diversity and landform development, thereby decreasing overall suitability in these areas. While mountainous regions to the west demonstrate high suitability. The MLP model further reveals that areas classified as very highly suitable for geomorphosites are dispersed across the western entirety of ZUW.

## Conclusions

This study primarily investigates the application of three topographic parameters (slope, slope length, and altitude) alongside three machine learning algorithms to identify potential geotourism sites. The findings indicate that this methodology is effective in discovering geomorphosites across extensive areas. However, it is essential to underscore that field visits are a vital component for verifying and exploring the geological features and landscapes of the region. These visits not only assess the scientific significance of these features but also provide detailed geological validation that models alone cannot offer.

This novel approach highlights the enhanced role of Machine learning in geotourism, contrasting with traditional methods that often rely on field visits and inventories, which can be challenging, especially in areas with unknown or inaccessible sites. The integration of machine learning techniques is recommended to diversify tourism offerings, particularly by showcasing geomorphosites of cultural significance that merit promotion through geotourism initiatives.

To fully realize the potential of these findings, it is essential to implement specific management measures. Conducting a systematic evaluation of geomorphosites will help identify sites with high tourism potential. Improving infrastructure, such as creating marked trails, informational panels, and observation points, will facilitate access to these sites. Additionally, awareness programs for local stakeholders and training for guides will ensure that visitors receive accurate and engaging interpretations of geological features.

Moreover, developing targeted promotional campaigns can attract a wider audience by highlighting the region's unique geological characteristics. Fostering partnerships among local stakeholders, authorities, and researchers will create synergies that enhance the effectiveness of these initiatives.

In conclusion, while this study supports the integration of machine learning in identifying potential geomorphosites, it also underscores the importance of traditional field assessments and collaborative efforts. By adopting these measures, we can not only boost geotourism but also contribute to the sustainable protection and valorization of the region's geological resources, ultimately enriching its geodiversity.

Finally, testing other machine learning algorithms, as well as integrating additional geomorphological factors in future studies or conducting comparative studies in other regions, will be very suitable.

**Acknowledgements** The authors are grateful for the Editor-in-Chief and the anonymous reviewers for their thorough reviews and constructive comments and they are very grateful to Mohamed Manaouch for the facilities and data provided during this study. The authors would like to thank Mr. N. Khan for his valuable help during the validation step.

**Author Contributions** LN and MM: geologic field investigation, inventory, characterization. MM and MA: classification and discussion. AJ and MH: manuscript preparation. QBP: revise and editing, supervision. MS and MM: geologic field investigation and data analysis. All authors read and approved the final version of the manuscript.

**Funding** The author(s) reported there is no funding associated with the work featured in this article.

**Data Availability** The data used to support the results of this research are available from the corresponding author upon request.

## Declarations

**Conflict of Interest** The authors declare no conflict of interest.

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